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FURTHER MATHEMATICS

9231/33

Paper 3 Further Mechanics

May/June 2023

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

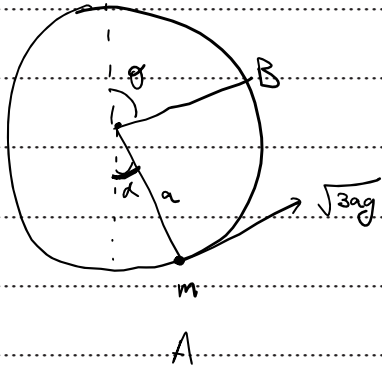
- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- 1 A particle P of mass m is attached to one end of a light inextensible string of length a . The other end of the string is attached to a fixed point O . The particle P is held at the point A , where OA makes an angle α with the downward vertical through O , and with the string taut. The particle P is projected perpendicular to OA in an upwards direction with speed $\sqrt{3ag}$. It then starts to move along a circular path in a vertical plane. The string goes slack when P is at B , where OB makes an angle θ with the upward vertical.

Given that $\cos \alpha = \frac{4}{5}$, find the value of $\cos \theta$.

[4]



Let u be speed @ bottom

@A:

$$\frac{mv^2}{r} = \frac{mu^2}{r} - 2mg + 2mg \cos(\alpha)$$

$$\frac{m(3ga)}{a} = \frac{mu^2}{a} - 2mg + \frac{8mg}{5}$$

$$3g = \frac{u^2}{a} - 2g + 1.6g$$

$$u^2 = 3.4ga$$

@B:

$$T = \frac{mv^2}{r} - 2mg + 3mg \cos(180 - \theta)$$

$$\frac{m(3.4ga)}{a} = 2mg - 3mg \cos(180 - \theta)$$

$$3.4 = 2 - 3 \cos(180 - \theta)$$

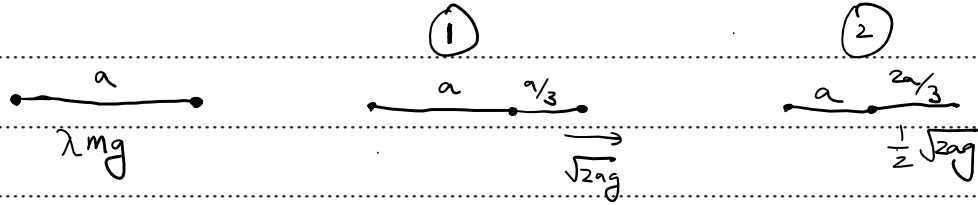
$$\cos(180 - \theta) = -\frac{7}{15}$$

$$\cos \theta = \frac{7}{15}$$

- 2 One end of a light elastic string, of natural length a and modulus of elasticity λmg , is attached to a fixed point O . The string lies on a smooth horizontal surface. A particle P of mass m is attached to the other end of the string. The particle P is projected in the direction OP . When the length of the string is $\frac{4}{3}a$, the speed of P is $\sqrt{2ag}$. When the length of the string is $\frac{5}{3}a$, the speed of P is $\frac{1}{2}\sqrt{2ag}$.

Find the value of λ .

[4]



Energy in ①:

$$\begin{aligned} & KE + EPE \\ &= \frac{1}{2}(2ag)(m) + \frac{(\lambda mg)(a/3)^2}{2a} \\ &= mga + \frac{\lambda mga}{18} \end{aligned}$$

Energy in ②:

$$\begin{aligned} & KE + EPE \\ &= \frac{1}{2}m\left(\frac{1}{2}\sqrt{2ag}\right)^2 + \frac{\lambda mg\left(\frac{2a}{3}\right)^2}{2a} \\ &= \frac{mga}{4} + \frac{2\lambda mga}{9} \\ &= \frac{mga}{4} + \frac{2\lambda mga}{9} \end{aligned}$$

↳ By CoE:

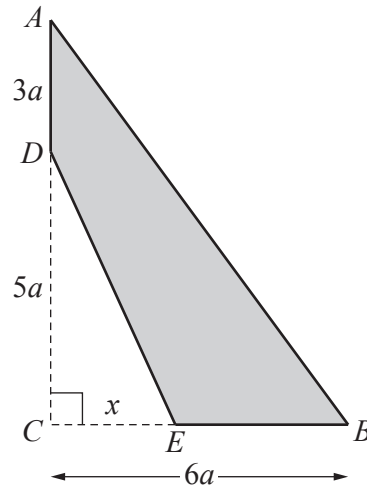
$$mga + \frac{\lambda mga}{18} = \frac{mga}{4} + \frac{2\lambda mga}{9}$$

$$1 + \frac{\lambda}{18} = \frac{1}{4} + \frac{2\lambda}{9}$$

$$18 + \lambda = 4.5 + 4\lambda$$

$$3\lambda = 13.5$$

$$\lambda = \frac{9}{2}$$



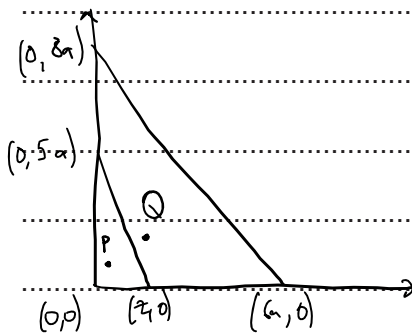
A uniform lamina is in the form of a triangle ABC , with $AC = 8a$, $BC = 6a$ and angle $ACB = 90^\circ$. The point D on AC is such that $AD = 3a$. The point E on CB is such that $CE = x$ (see diagram). The triangle CDE is removed from the lamina.

- (a) Find, in terms of a and x , the distance of the centre of mass of the remaining object $ADEB$ from AC . [4]

↳ find $\bar{x}$

$$P: \left(\frac{x}{3}, \frac{5a}{3} \right)$$

$$Q: \left(2a, \frac{8a}{3} \right)$$



Taking moments about AC:

$$(\bar{x}) \left(\frac{6a}{2} \right) \left(\frac{8a}{2} - \frac{5ax}{2} \right) = \frac{6a}{2} \left(\frac{8a}{2} \right) (2a) + \frac{5ax}{2} \left(\frac{x}{3} \right)$$

$$\bar{x} \left(\frac{9a}{2} \right) (48a - 5x) = \frac{9a}{2} (96a^2 - \frac{5x^2}{3})$$

$$\bar{x} = \frac{288a^2 - 5x^2}{144a - 15x}$$

The object $ADEB$ is on the point of toppling about the point E when the object is in the vertical plane with its edge EB on a smooth horizontal surface.

(b) Find x in terms of a .

[3]

$$\frac{288a^2 - 5x^2}{144a - 15x} = x$$

$$288a^2 - 5x^2 = 144ax - 15x^2$$

$$10x^2 - 144ax + 288a^2 = 0$$

$$5x^2 - 72xa + 144a^2 = 0$$

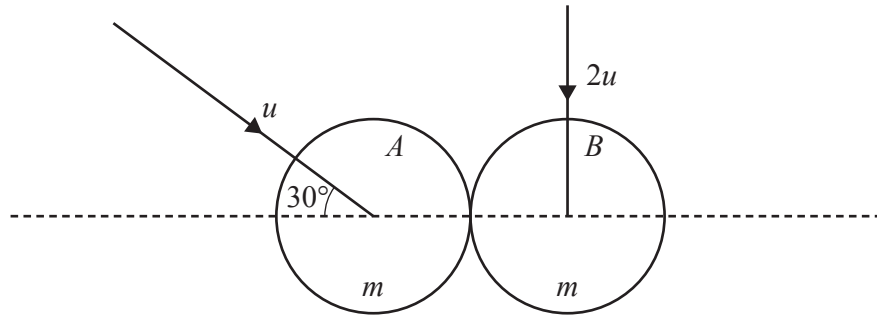
$$x = \frac{72a \pm \sqrt{5184a^2 - 4(5)(144a^2)}}{10}$$

$$= 7.2a \pm 4.8a$$

$$= 12a \text{ or } 2.4a$$

$$\downarrow$$

$$(as x \leq 6a)$$



Two identical smooth uniform spheres A and B each have mass m . The two spheres are moving on a smooth horizontal surface when they collide with speeds u and $2u$ respectively. Immediately before the collision, A 's direction of motion makes an angle of 30° with the line of centres, and B 's direction of motion is perpendicular to the line of centres (see diagram). After the collision, A and B are moving in the same direction. The coefficient of restitution between the spheres is e .

(a) Find the value of e .

[5]

→ Only horizontal components are affected by collisions:

$$m \cdot u \cos 30^\circ$$

$$m u \cos 30^\circ = m v_{Ax} + m v_{Bx}$$

$$\frac{\sqrt{3}u}{2} = v_{Ax} + v_{Bx} \quad (1)$$

$$\frac{v_{Bx} - v_{Ax}}{\frac{\sqrt{3}u}{2}} = e$$

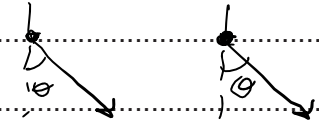
$$v_{Bx} - v_{Ax} = \frac{\sqrt{3}eu}{2} \quad (2)$$

Adding (1) & (2):

$$2v_{Bx} = \frac{\sqrt{3}u}{2}(e+1)$$

$$v_{Bx} = \frac{\sqrt{3}u}{4}(e+1)$$

$$v_{Ax} = \frac{\sqrt{3}u}{4}(1-e)$$



$$\frac{v_{Ax}}{v_{Ay}} = \frac{v_{Bx}}{v_{By}}$$

$$\frac{\left(\frac{\sqrt{3}u}{4}\right)(1-e)}{u \sin 30^\circ} = \frac{\left(\frac{\sqrt{3}u}{4}\right)(1+e)}{2u}$$

$$\frac{\sqrt{3}}{2}(1-e) = \frac{\sqrt{3}}{8}(1+e)$$

$$4-4e = e+1$$

$$3 = 5e$$

$$e = \frac{3}{5}$$

(b) Find the loss in the total kinetic energy of the spheres as a result of the collision.

[3]

Initially:

$$KE = \frac{1}{2}mv^2 + \frac{1}{2}m(2v)^2 = \frac{5}{2}mv^2$$

Finally:

$$V_A = \sqrt{\left(\frac{v}{2}\right)^2 + \left(\left(\frac{\sqrt{3}v}{4}\right)\left(\frac{2}{5}\right)\right)^2} \quad \left(\frac{3}{4}\right)\left(\frac{4}{25}\right)$$

$$= \sqrt{\frac{v^2}{4} + \frac{3v^2}{100}} = \sqrt{\frac{28v^2}{100}} = \sqrt{\frac{7v^2}{25}}$$

$$V_B = \sqrt{(2v)^2 + \left(\left(\frac{\sqrt{3}v}{4}\right)\left(\frac{8}{5}\right)\right)^2} \quad \left(\frac{3}{4}\right)\left(\frac{4}{25}\right)$$

$$= \sqrt{4v^2 + \frac{12v^2}{25}} = \frac{48}{25}$$

$$= \sqrt{\frac{112v^2}{25}}$$

$$\text{Final KE} = \frac{1}{2}m\left(\frac{112v^2}{25}\right) + \frac{1}{2}m\left(\frac{7v^2}{25}\right)$$

$$= \frac{119}{50}mv^2$$

$$\text{Loss} = \frac{125}{50}mv^2 - \frac{119}{50}mv^2$$

$$= \frac{3}{25}mv^2$$

- 5 One end of a light elastic string, of natural length $12a$ and modulus of elasticity kmg , is attached to a fixed point O . The other end of the string is attached to a particle of mass m . The particle moves with constant speed $\frac{3}{2}\sqrt{3ag}$ in a horizontal circle with centre at a distance $12a$ below O . The string is inclined at an angle θ to the downward vertical through O .

(a) Find, in terms of a , the extension of the string.

[5]

$$T \cos \theta = mg$$

$$T = \frac{mg}{\left(\frac{12a}{12a+x}\right)} = \frac{mg(12a+x)}{12a}$$

$$r = \sqrt{(12a+x)^2 - (12a)^2}$$

$$= \sqrt{x^2 + 24ax}$$

$$T \sin \theta = \frac{mv^2}{r}$$

$$\left(\frac{mg(12a+x)}{12a}\right) \left(\frac{\sqrt{x^2 + 24ax}}{12a+x}\right) = \frac{m \left(\frac{9}{4}\right) (3ag)}{\sqrt{x^2 + 24ax}}$$

$$\frac{mg(x^2 + 24ax)}{12a \cdot 3} = \frac{27}{4} mga$$

$$x^2 + 24ax = 81a^2$$

$$x^2 + 24ax - 81a^2$$

$$(x+27a)(x-3a) = 0$$

$$\hookrightarrow x = 3a \text{ or } -27a$$

$$\hookrightarrow \text{not possible}$$

$$\boxed{x = 3a}$$

(b) Find the value of k .

[3]

$$T = \frac{\lambda x}{l}$$

$$\frac{mg(12a+x)}{12a} = \frac{kmg(3a)}{12a}$$

$$\frac{15mg}{12} = \frac{3kmg}{12}$$

$$\boxed{k=5}$$

- 6 A particle of mass m kg falls vertically under gravity, from rest. At time t s, P has fallen x m and has velocity v ms⁻¹. The only forces acting on P are its weight and a resistance of magnitude $kmgv$ N, where k is a constant.

(a) Find an expression for v in terms of t , g and k .

[5]

$$a_{\text{down}} = g - kgv$$

$$\frac{dv}{dt} = g - kgv$$

$$\frac{dv}{1 - kv} = g dt$$

$$\int \frac{1}{1 - kv} dv = \int g dt$$

$$\frac{\ln|1 - kv|}{-k} = gt + c$$

$\Rightarrow$ starts @ rest, so
when $t=0$, $v=0$

$$\frac{\ln 1}{-k} = c = 0$$

$$\frac{\ln|1 - kv|}{-k} = gt$$

$$\ln|1 - kv| = -gtk$$

$$\ln|1 - kv| = -gtk$$

$$1 - kv = e^{-gtk}$$

$$\frac{1 - e^{-gtk}}{k} = v$$

- (b) Given that $k = 0.05$, find, in metres, how far P has fallen when its speed is 12 m s^{-1} . [5]

$$v = 20 - 20e^{-0.05gt}$$

when $v = 12$

$$8 = 20e^{-0.05gt}$$

$$e^{0.05gt} = \frac{5}{2}$$

$$t = \frac{\ln 2.5}{0.05g} = 1.833 \text{ s}$$

$$x = \int_0^{1.833} (20 - 20e^{-0.05gt}) dt$$

$$= \left[20t + \frac{20e^{-0.05gt}}{0.05g} \right]_0^{1.833}$$

$$= 20(1.833) + \frac{20(\frac{2}{5})}{0.05g} - \frac{20e^0}{0.05g}$$

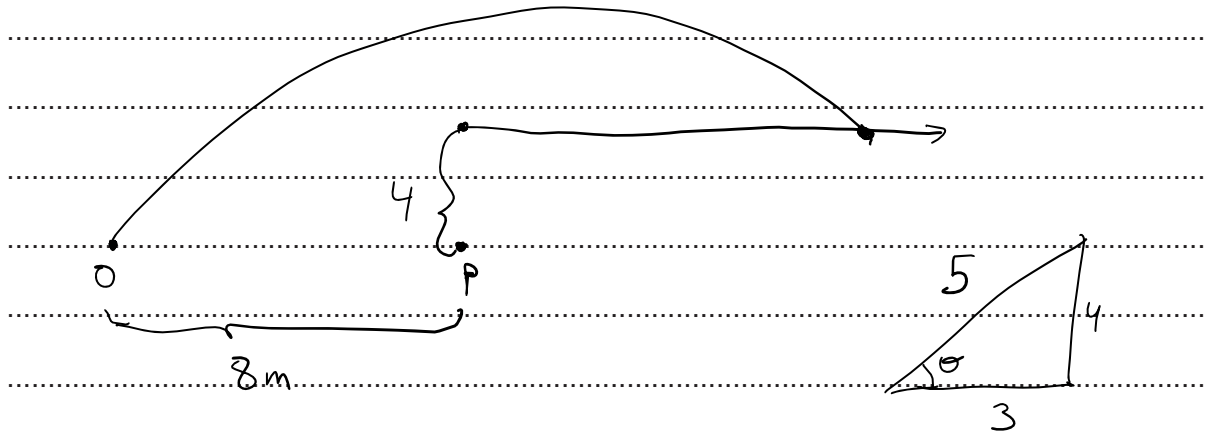
$$= 36.65 + \frac{8(20)}{g} - 40$$

$$\boxed{= 12.7 \text{ m}}$$

- 7 The points O and P are on a horizontal plane, a distance 8 m apart. A ball is thrown from O with speed $u \text{ m s}^{-1}$ at an angle θ above the horizontal, where $\tan \theta = \frac{4}{3}$. At the same instant, a model aircraft is launched with speed 5 m s^{-1} parallel to the horizontal plane from a point 4 m vertically above P . The model aircraft moves in the same vertical plane as the ball and in the same horizontal direction as the ball. The model aircraft moves horizontally with a constant speed of 5 m s^{-1} . After T s, the ball and the model aircraft collide.

(a) Find the value of T .

[6]



@ Time T :

Ball: Horizontal distance = $5T$

Plane: Horizontal distance = $5T \cos \theta = \frac{30T}{5}$

Ball: Height = 4 m

Plane: $y = x \tan \theta - \frac{g x^2}{2u^2} \sec^2 \theta$

$$= \frac{4}{3}x - \frac{125x^2}{9u^2}$$

$$\hookrightarrow = \frac{40T}{5} - 5T^2$$

$$\frac{40T}{5} - 5T^2 = 4$$

$$5T + 8 = \frac{30T}{5}$$

$$\frac{5T^2 + 4}{4} = \frac{5T + 8}{3}$$

$$\sin \theta = \frac{4}{5}$$

$$\cos \theta = \frac{3}{5}$$

$$\tan \theta = \frac{4}{3}$$

$$15T^2 + 12 = 20T + 32$$

$$15T^2 - 20T - 20 = 0$$

$$3T^2 - 4T - 4 = 0$$

$$(3T + 2)(T - 2)$$

$$\hookrightarrow \boxed{T = 2}$$

(b) Find the direction in which the ball is moving immediately before the collision.

[3]

$$5\tau + 8 = \frac{30\tau}{5}$$

$$1.8 = \frac{6v}{5}$$

$$v = 15 \text{ ms}^{-1}$$

for θ from the horizontal:

$$\tan \theta = \frac{v \sin \theta - gt}{v \cos \theta}$$

$$= \frac{15\left(\frac{4}{5}\right) - 20}{15\left(\frac{3}{5}\right)} = \frac{-8}{9}$$

$$\theta = 41.6^\circ \text{ below horizontal}$$

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.

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